

EKLAKH DEWAN

Data Scientist — Machine Learning — NLP — Statistical Analysis

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SUMMARY

Data Scientist focused on extracting actionable insights from structured and unstructured data using statistical analysis, machine learning, and NLP. Experienced in exploratory data analysis, feature engineering, similarity modeling, classification, embeddings, data preprocessing, and evaluation using Python, SQL, Pandas, NumPy, and scikit-learn.

EDUCATION

KPRIET

B.Tech. in Artificial Intelligence & Data Science

2027

CGPA: 8.55

EXPERIENCE

Artificial Intelligence Intern

Flowrage Technology

Virtual Internship — 4 Weeks

- Processed unstructured resume and job-description data for AI-assisted matching and recommendation workflows.
- Applied **TF-IDF** and dense semantic embeddings to engineer textual features and represent candidate-job relationships.
- Developed matching and ranking logic using similarity-based methods and evaluated outputs to refine recommendation quality.
- Integrated **Google Gemini API** into application workflows for AI-assisted text processing and recommendation functionality.

SELECTED DATA SCIENCE PROJECTS

Insurance Claims ML Automation

[GitHub](#)

Python, Pandas, scikit-learn, XGBoost, Statistical Analysis

- Designed an end-to-end analytics and machine learning workflow covering data ingestion, cleaning, feature engineering, classification, and model evaluation for insurance claims.
- Analyzed historical claims data to identify patterns associated with invalid or potentially fraudulent claims and addressed class imbalance during model development.
- Evaluated predictive models using precision and recall to examine the trade-off between correctly identifying anomalous claims and limiting false positives.

AI-Powered Job Recommendations

[GitHub](#)

Python, Pandas, scikit-learn, SentenceTransformers, Streamlit

- Conducted text preprocessing and feature engineering on resumes and job descriptions using TF-IDF and dense semantic embeddings.
- Used cosine similarity and semantic matching to rank job opportunities against candidate profiles and identify relevant skill gaps.
- Developed an interactive Streamlit dashboard to present ranked recommendations and make matching signals interpretable.

Enterprise RAG Evaluation Pipeline

[GitHub](#)

Python, FAISS, BM25, SentenceTransformers, Retrieval Evaluation

- Developed an evaluation pipeline for comparing dense, sparse, hybrid, and reranked retrieval strategies over unstructured document data.
- Measured retrieval quality using **Recall@K**, **MRR**, and **nDCG** to evaluate ranking effectiveness across retrieval stages.
- Analyzed retrieval behavior using semantic embeddings and cross-encoder reranking to improve relevance of evidence supplied to downstream generation.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript

Data Science: Pandas, NumPy, scikit-learn, EDA, Feature Engineering, Data Cleaning, Statistical Analysis

Machine Learning: Classification, Predictive Modeling, XGBoost, Model Evaluation, Similarity Modeling

NLP: TF-IDF, SentenceTransformers, Embeddings, Semantic Similarity, Text Classification, NLP Preprocessing

Retrieval: FAISS, BM25, Hybrid Search, RRF, Cross-Encoder Reranking, Recall@K, MRR, nDCG

Data & Application: PostgreSQL, FastAPI, Streamlit, GitHub Actions

ANALYTICAL FOCUS

Exploratory Data Analysis — Feature Engineering — Statistical Testing — Predictive Modeling — NLP — Similarity Analysis — Model Evaluation — Data-Driven Decision Making